

11 Eigenvalue Decomposition

11.3 Eigendecomposition of Non-symmetric Matrices

V is no longer orthogonal



Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master in This Section

1

EVD of a non-symmetric matrix: V is no longer an orthogonal matrix

2

V and its inverse cancel each other's effect; their determinants multiply to 1

3

This section decomposes V into a shear operation and a scale operation

4

The EVD of a non-symmetric matrix reads as “scale $\rightarrow$ shear $\rightarrow$ scale $\rightarrow$ shear $\rightarrow$ scale”

5

Uniform scaling matrix: every direction is an eigen-direction

6

Non-uniform scaling: each direction has its own stretch factor

7

Rotation matrices: usually cannot be diagonalized over the reals — complex numbers appear

8

A zero eigenvalue means a dimension is lost in the transformation

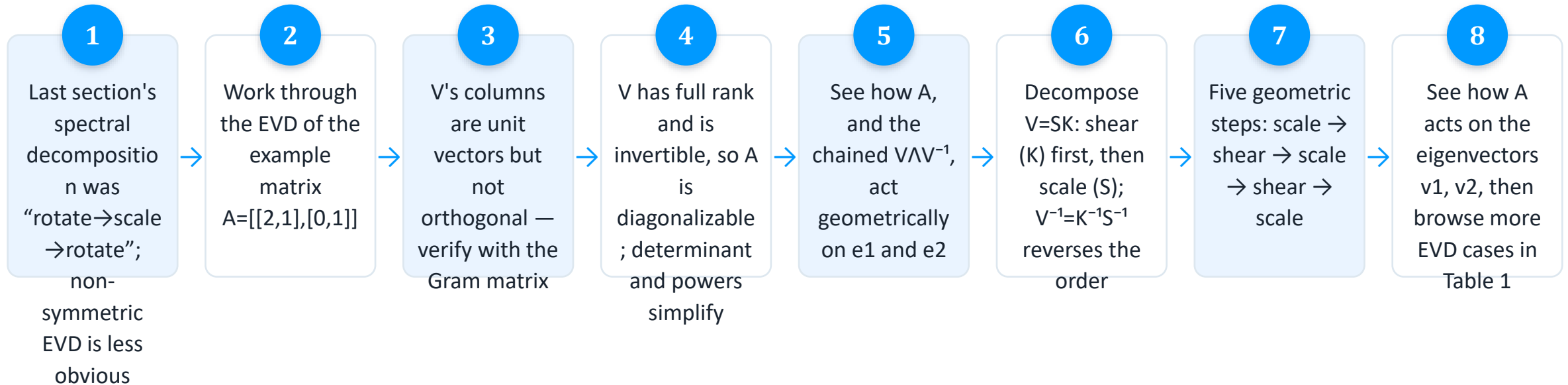
9

A negative eigenvalue means a reflection

10

A matrix with eigenvalues 0 and 1 can still be diagonalizable

From $A=V\Lambda V^{-1}$ to decomposing V , then more EVD examples



The eigendecomposition of a non-symmetric square matrix A is still written as

$$A = V\Lambda V^{-1}$$

- The columns of V are eigenvectors of A , but V is generally not orthogonal, so $V^{-1} \neq V^T$
- Λ is diagonal, with A 's eigenvalues on the main diagonal — same as spectral decomposition

Given the non-symmetric matrix A , its eigendecomposition is

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -\sqrt{2}/2 \\ 0 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & \sqrt{2} \end{bmatrix} = V\Lambda V^{-1}$$

- V 's columns $v_1 = [1, 0]^T$ and $v_2 = [-\sqrt{2}/2, \sqrt{2}/2]^T$ are unit vectors, but not orthogonal
- The rightmost matrix is V 's inverse; since V is not orthogonal, V^{-1} cannot be written as V^T

Compute the Gram matrix of V , $V^T V$

$$V^T V = V^T [v_1, v_2]$$

- Write V as $[v_1, v_2]$; expanding the matrix product gives all four entries of $V^T V$
- Diagonal entries are the squared lengths of v_1, v_2 ; off-diagonal entries are their inner product

The computed result of the Gram matrix $V^T V$

$$V^T V = \begin{bmatrix} 1 & -\sqrt{2}/2 \\ -\sqrt{2}/2 & 1 \end{bmatrix}$$

- Diagonal entries equal 1, confirming v_1 and v_2 are unit vectors
- Off-diagonal entries are nonzero, so v_1 and v_2 are not orthogonal — the angle between them is 135°

The inverse matrix of V is

$$V^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & \sqrt{2} \end{bmatrix}$$

- The columns of V^{-1} , $[1,0]^T$ and $[1,\sqrt{2}]^T$, are not both unit vectors
- This further confirms V is not orthogonal — for an orthogonal matrix, the inverse equals the transpose

DIAGONALIZABLE

V 's columns are linearly independent, so V is full rank and invertible — A is diagonalizable

$$V^{-1}AV = \Lambda = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

- V being invertible is the necessary and sufficient condition — orthogonality of V is irrelevant
- Once a matrix is successfully diagonalized, determinant and power computations simplify a lot

The determinant reduces to a product of eigenvalues; matrix powers simplify too

$$\det(A) = \lambda_1 \lambda_2 = 2, \quad A^2 = V\Lambda^2V^{-1}, \quad A^n = V\Lambda^nV^{-1}$$

- The determinant no longer needs cofactor expansion — just multiply the diagonal entries of Λ
- Next, a geometric view helps us understand A and its eigendecomposition

Figure 1 · The Linear Transformation of A on e_1, e_2

- $Ae_1=[2,0]^T$, $Ae_2=[1,1]^T$; the unit square is stretched and sheared into a parallelogram

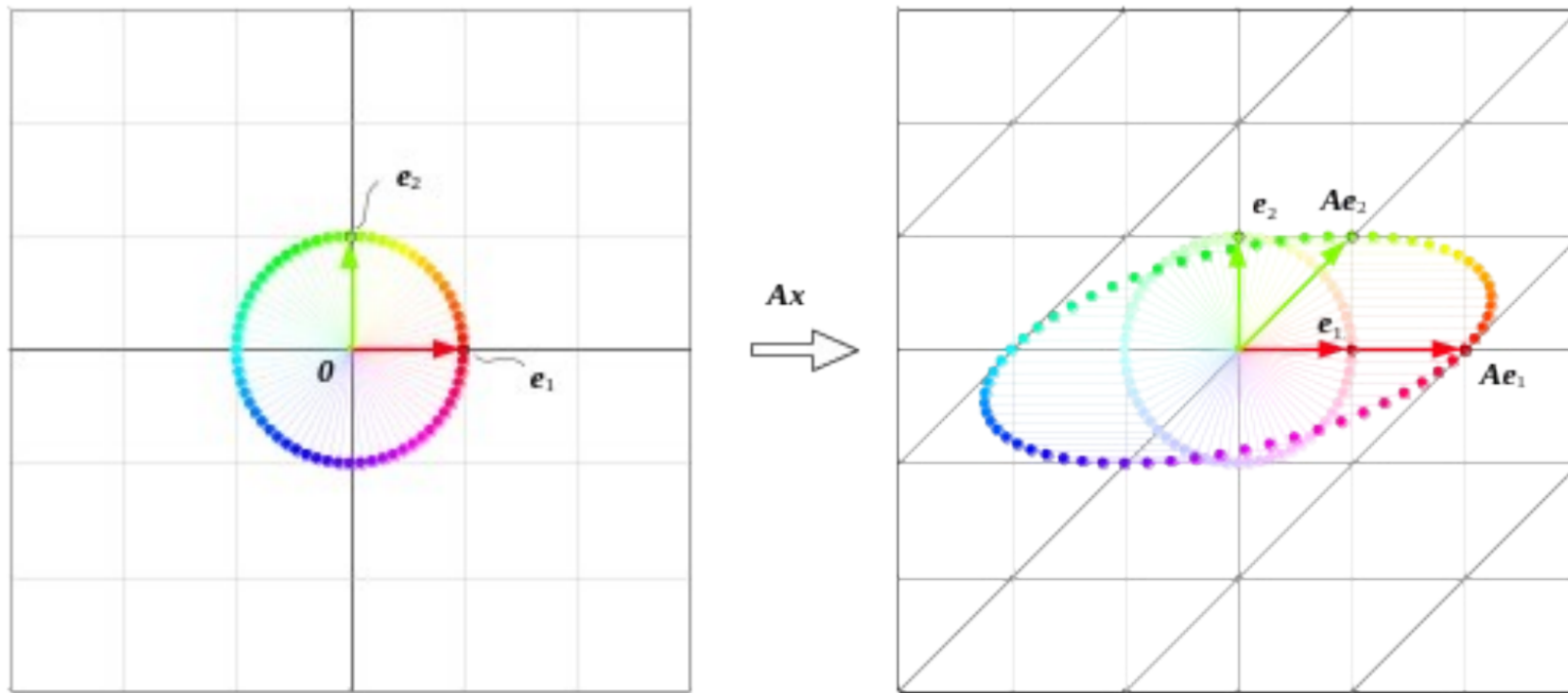
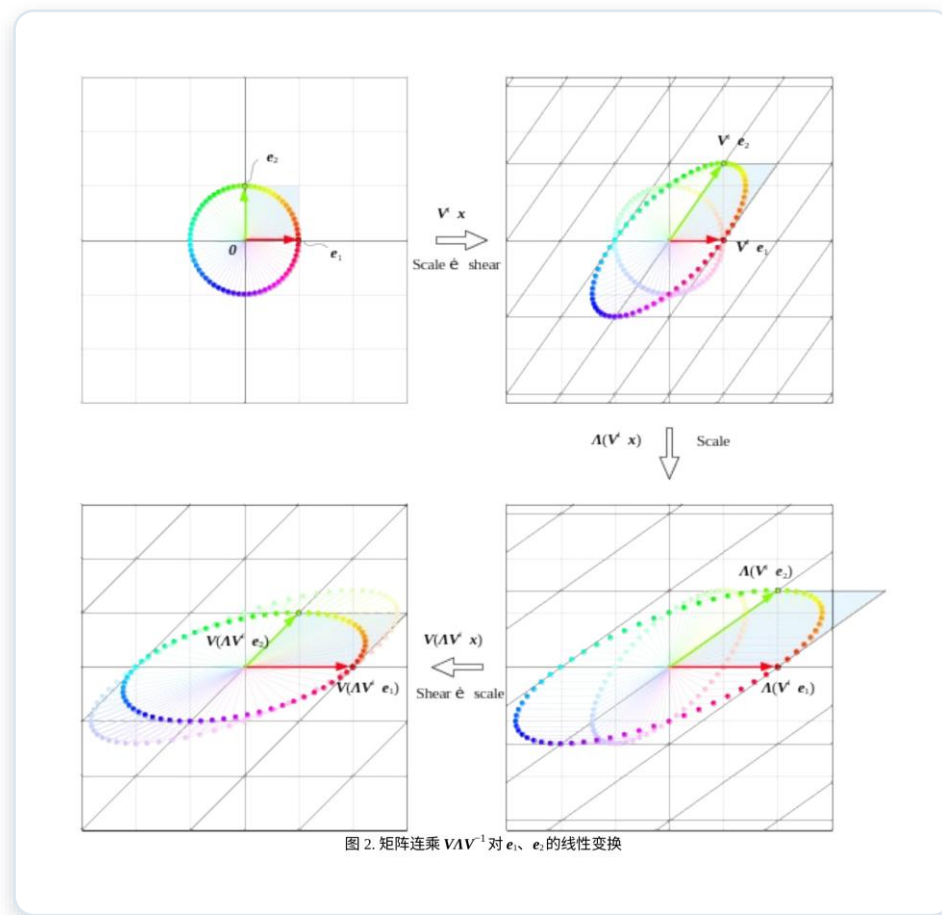


图 1. 矩阵 A 对 e_1, e_2 的线性变换

Figure 2 · The Chained Transformation $V\Lambda V^{-1}$ on e_1, e_2

- $V^{-1}x$ (scale+shear) $\rightarrow \Lambda(V^{-1}x)$ (scale) $\rightarrow V(\Lambda V^{-1}x)$ (shear+scale) $= Ax$



Looking closely at V, it factors into two matrices

$$V = SK = \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} 1 & -\sqrt{2}/2 \\ 0 & 1 \end{bmatrix}$$

- K shears along the horizontal axis; S scales — more precisely, along the vertical axis
- $Vx=SKx$ applies to x in this order: shear (K) first, then scale (S)

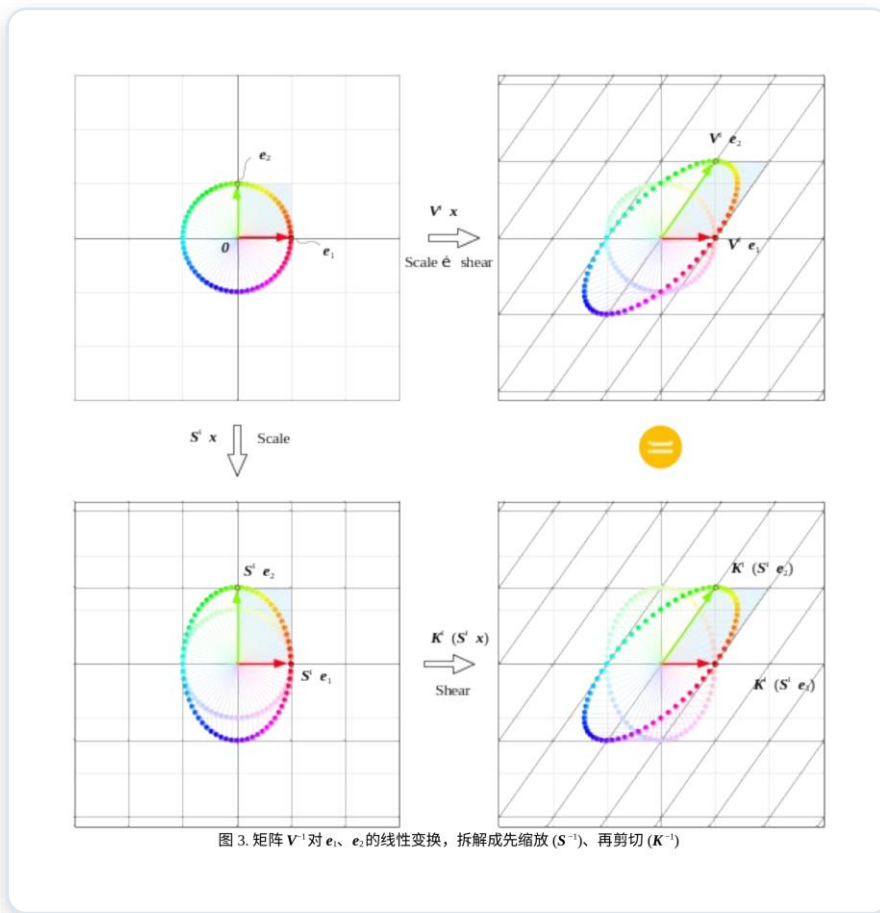
Correspondingly, the inverse of V can be written as

$$V^{-1} = K^{-1}S^{-1} = \begin{bmatrix} 1 & \sqrt{2}/2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{2} \end{bmatrix}$$

- $V^{-1}x = K^{-1}S^{-1}x$ — the order reverses V 's: scale (S^{-1}) first, then shear (K^{-1})

Figure 3 · V^{-1} on e_1, e_2 , Decomposed into Scale then Shear

- V^{-1} acts as scaling by S^{-1} first, then shearing by K^{-1}



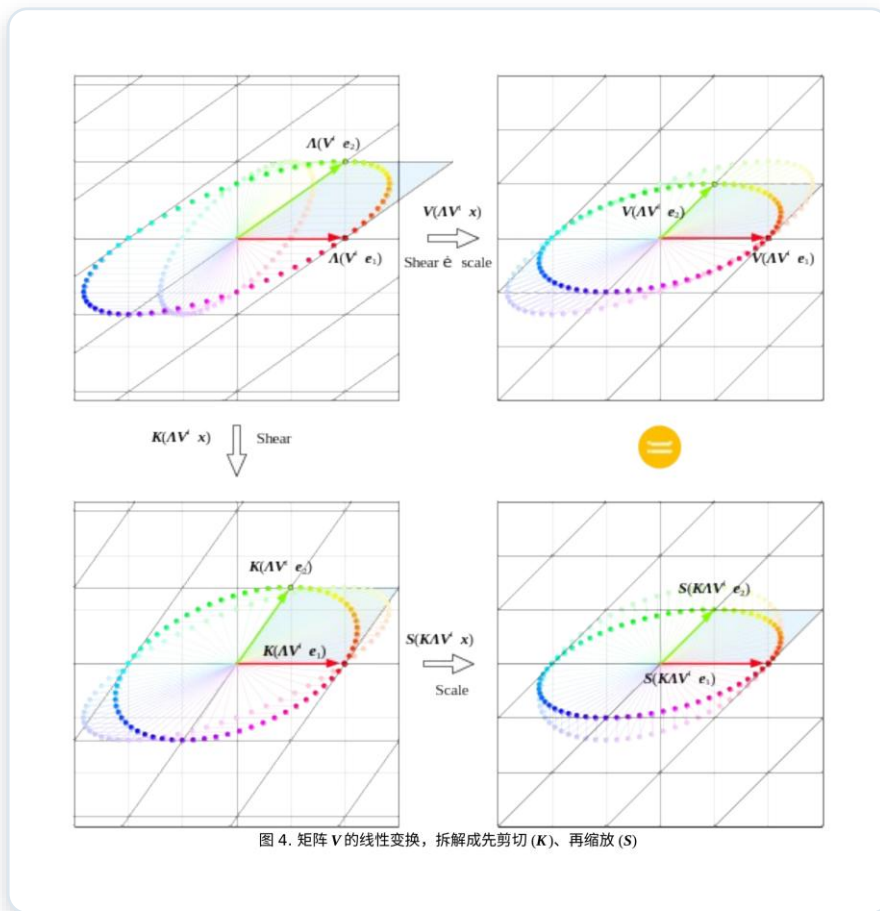
The determinant of the inverse matrix V^{-1} is

$$\det(V^{-1}) = \sqrt{2}$$

- This means the parallelogram grid area in Figure 3 grows by a factor of $\sqrt{2}$
- Shear K^{-1} never changes area — it is scale S^{-1} that changes the area here

Figure 4 · V on e_1, e_2 , Decomposed into Shear then Scale

- V acts oppositely to V^{-1} : shear (K) first, then scale (S)



The determinant of V, and its product with $\det(V^{-1})$, which is always 1

$$\det(V) = \sqrt{2}/2, \quad \det(V^{-1}) \cdot \det(V) = 1$$

- $\det(V)$ means the area shrinks to $\sqrt{2}/2$ of the original
- The determinants of a matrix and its inverse always multiply to 1 — V^{-1} and V cancel each other's area effect

The area change caused by A comes entirely from $\det(\Lambda)$

$$\det(A) = \det(\Lambda) = \lambda_1 \lambda_2$$

- V^{-1} and V cancel out their effect on area — only the diagonal Λ truly changes the area

SCALE → SHEAR → SCALE → SHEAR → SCALE

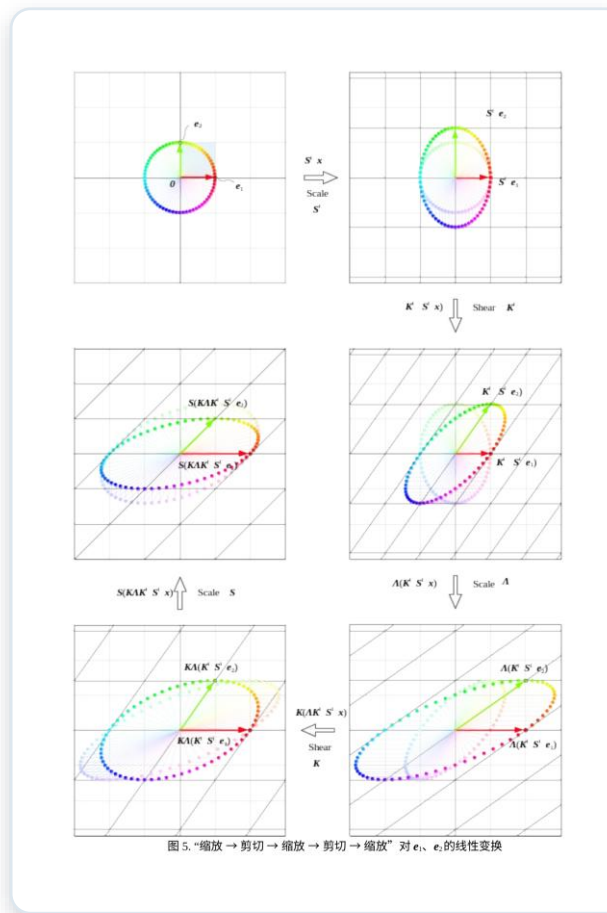
The eigendecomposition of A unfolds into five geometric steps

$$A = SK\Lambda K^{-1}S^{-1}$$

- Applied to vector x in order: scale (S^{-1}) → shear (K^{-1}) → scale (Λ) → shear (K) → scale (S)

Figure 5 · “Scale → Shear → Scale → Shear → Scale” on e_1, e_2

- The five arrows correspond to $S^{-1}, K^{-1}, \Lambda, K, S$ — chained together they equal Ax



A's action on the eigenvectors v_1, v_2 corresponds to

$$Av_1 = 2v_1, \quad Av_2 = 1 \times v_2$$

- $\lambda_1=2, v_1=[1,0]^T=e_1; \lambda_2=1, v_2=[-\sqrt{2}/2, \sqrt{2}/2]^T$
- A's action on an eigenvector is the simplest possible: only length scales, direction stays the same

Figure 6 · The Linear Transformation of A on v_1, v_2

- Av_1 stretches 2× along v_1 ; Av_2 coincides with v_2 , unchanged in length ($\lambda_2=1$)

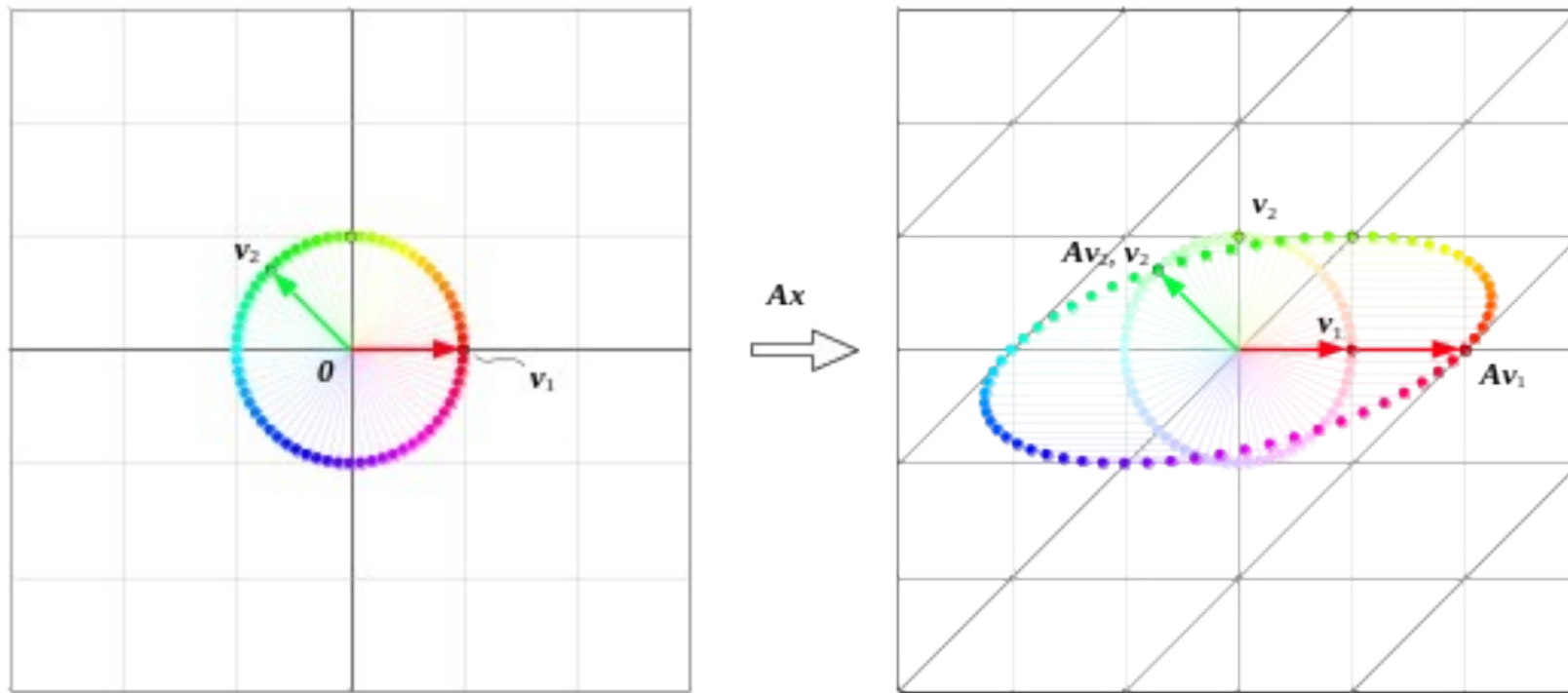
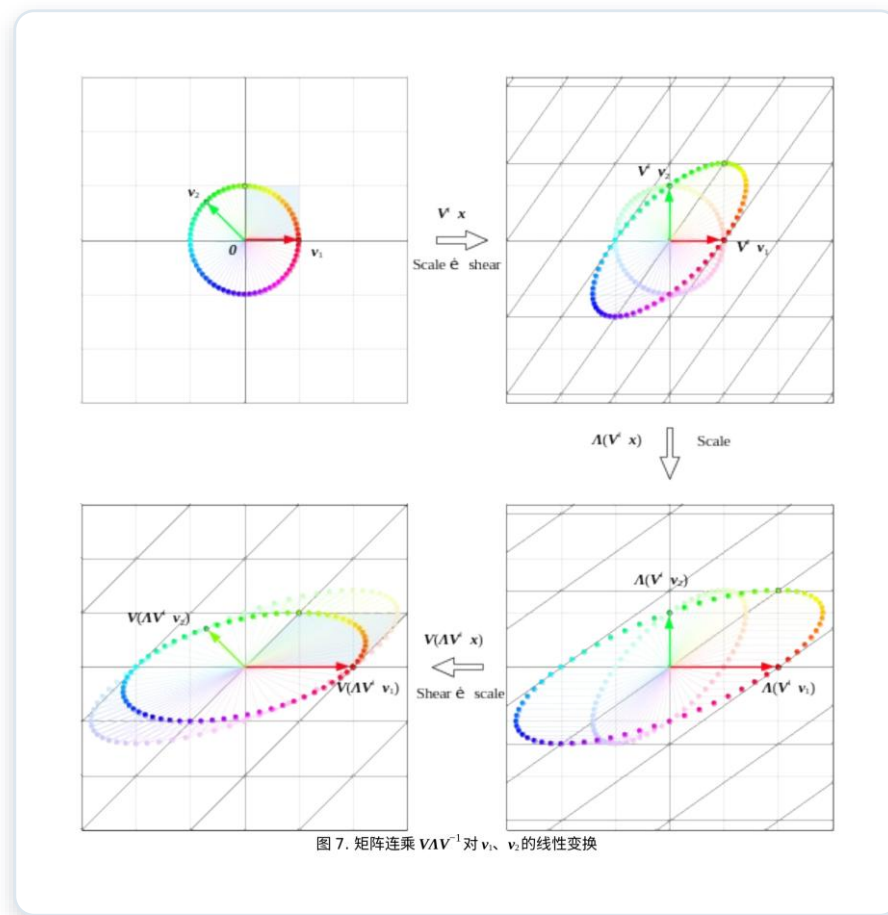


图 6. 矩阵 A 对 v_1 、 v_2 的线性变换

Figure 7 · The Chained Transformation $V\Lambda V^{-1}$ on v_1, v_2

- The steps mirror Figure 2 exactly, just applied to v_1, v_2 — try analyzing it yourself



The Example Above Is Just One Case of Non-symmetric EVD

- ① Uniform scaling (e.g. $\text{diag}(2,2)$): every nonzero vector is an eigenvector, naturally diagonalizable
- ② Non-uniform scaling (e.g. $\text{diag}(1/2,2)$): the axis directions remain eigen-directions with different stretch factors
- ③ Rotation matrices: eigenvalues/eigenvectors are generally complex — not diagonalizable over the reals
- ④ Shear matrices: often have a repeated eigenvalue with only one independent direction — not diagonalizable
- ⑤ Rank-deficient matrices: a zero eigenvalue collapses that direction to a point — a dimension is lost
- ⑥ Matrices with a negative eigenvalue: vectors along that direction get reversed — a reflection
- ⑦ Symmetric matrices: always diagonalizable, even with special eigenvalues like 0 and 1

TABLE 1 · MORE EIGENDECOMPOSITIONS (1/3)

Uniform Scaling, Non-uniform Scaling, and Rotation

Matrix A	Eigenvalues, Eigenvectors	Diagonalizable	Note
$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$	$\lambda_{1,2}=2$; any nonzero vector can serve as an eigenvector	Yes (already diagonal)	Uniform scaling; a circle becomes a bigger circle, shape unchanged
$\begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix}$	$\lambda_1=1/2, v_1=[1,0]$; $\lambda_2=2, v_2=[0,1]$	Yes (already diagonal)	Non-uniform scaling; a circle becomes a vertical ellipse
$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$	$\lambda_{1,2}=\cos\theta \pm i\sin\theta$; $v_1=[1,-i], v_2=[1,i]$	Only if $\sin\theta=0$, else complex	Rotation matrix; generally cannot be diagonalized over the reals
$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	$\lambda_{1,2}=1$ (repeated); $v_1=[1,0], v_2=[-1,0]$ are collinear	No	A shear; only one independent eigen-direction
$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$	$\lambda_{1,2}=1$ (repeated); $v_1=[0,1], v_2=[0,-1]$ are collinear	No	A shear; only one independent eigen-direction
$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	$\lambda_{1,2}=1$ (repeated); $v_1=[0,1], v_2=[0,1]$ are collinear	No	A shear; only one independent eigen-direction

TABLE 1 · MORE EIGENDECOMPOSITIONS (2/3)

Shear, Reflection, and Rank-one Matrices

Matrix A	Eigenvalues, Eigenvectors	Diagonalizable	Note
$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$	$\lambda_{1,2}=1$ (repeated); $v_1=[1,0]$, $v_2=[1,0]$ are collinear	No	A shear; only one independent eigen-direction
$\begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$	$\lambda_{1,2}=-1$ (repeated); $v_1=[1,0]$, $v_2=[-1,0]$ are collinear	No	A negative eigenvalue means reflection, combined with shear
$\begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$	$\lambda_1=1, v_1=[1,0]$; $\lambda_2=-1, v_2=[-\sqrt{5}/5, 2\sqrt{5}/5]$	Yes	One positive, one negative eigenvalue — half stays, half reflects
$\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$	$\lambda_1=0, v_1=[\sqrt{2}/2, \sqrt{2}/2]$; $\lambda_2=1, v_2=[-\sqrt{2}/2, \sqrt{2}/2]$	Yes	Rank-one matrix; the 0-eigenvalue direction collapses to a point
$\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$	$\lambda_1=0, v_1=[1,0]$; $\lambda_2=1, v_2=[\sqrt{2}/2, \sqrt{2}/2]$	Yes	Rank-one matrix; the column space degenerates to a line
$\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$	$\lambda_1=0, v_1=[0,1]$; $\lambda_2=1, v_2=[\sqrt{2}/2, \sqrt{2}/2]$	Yes	Rank-one matrix; the column space degenerates to a line

TABLE 1 · MORE EIGENDECOMPOSITIONS (3/3)

Rank-one and Symmetric Matrices

Matrix A	Eigenvalues, Eigenvectors	Diagonalizable	Note
[[0,1],[0,-1]]	$\lambda_1=0, v_1=[1,0]; \lambda_2=-1, v_2=[-\sqrt{2}/2, \sqrt{2}/2]$	Yes	Rank-one matrix, combining dimension loss and reflection
[[1,1],[0,0]]	$\lambda_1=1, v_1=[0,1]; \lambda_2=0, v_2=[-\sqrt{2}/2, \sqrt{2}/2]$	Yes	Rank-one matrix; the column space degenerates to a line
[[0,0],[1,1]]	$\lambda_1=1, v_1=[0,1]; \lambda_2=0, v_2=[\sqrt{2}/2, -\sqrt{2}/2]$	Yes	Rank-one matrix; the column space degenerates to a line
[[1,1],[1,1]]	$\lambda_1=2, v_1=[\sqrt{2}/2, \sqrt{2}/2]; \lambda_2=0, v_2=[-\sqrt{2}/2, \sqrt{2}/2]$	Yes	Symmetric matrix; eigenvectors are naturally orthogonal
[[-1,1],[1,-1]]	$\lambda_1=0, v_1=[\sqrt{2}/2, \sqrt{2}/2]; \lambda_2=-2, v_2=[-\sqrt{2}/2, \sqrt{2}/2]$	Yes	Symmetric matrix; eigenvectors are naturally orthogonal
[[0,1],[1,0]]	$\lambda_1=1, v_1=[\sqrt{2}/2, \sqrt{2}/2]; \lambda_2=-1, v_2=[-\sqrt{2}/2, \sqrt{2}/2]$	Yes	Symmetric matrix; reflection across $y=x$

A Few Empirical Rules for Non-symmetric EVD

① Diagonal matrices: eigenvectors are the axis directions; any nonzero vector works for uniform scaling

② Repeated eigenvalue + shear structure: usually only one eigen-direction, matrix not diagonalizable

③ A zero eigenvalue: the matrix is rank-deficient, and information along that direction is lost

④ A negative eigenvalue: vectors along that direction get reversed — geometrically, a reflection

⑤ Symmetric matrices: always diagonalizable regardless of zero or repeated eigenvalues, with orthogonal eigenvectors

⑥ Rotation matrices: eigenvalues are generally complex — only real-diagonalizable when $\sin\theta=0$

Key Terms in This Section

Non-symmetric EVD $A=V\Lambda V^{-1}$, where V 's columns are eigenvectors but V is generally not orthogonal

Diagonalizable When V has full rank, $\Lambda=V^{-1}AV$ — independent of whether V is orthogonal

Scale Matrix S Stretches or compresses along axis directions; changes area

Rank-one Matrix Has a zero eigenvalue; that direction collapses to a point — a dimension is lost

Complex Eigenvalue Typical of rotation matrices; cannot be diagonalized over the reals

Gram Matrix VTV Used to check whether V 's columns are unit vectors and orthogonal

Shear Matrix K Translates along one axis direction; never changes area

Determinant Product The determinants of a matrix and its inverse always multiply to 1

Negative Eigenvalue Vectors along that direction get reversed — geometrically, a reflection

Table 1 18 example 2×2 matrices covering scaling, rotation, shear, reflection, and rank deficiency

Further Reading

Review · 11.2 Spectral Decomposition

EVD of a real symmetric matrix — “rotate→scale→rotate”, in contrast with this section

Further Reading · 11.4 Gram Matrix

The VTV used here is one of the most common symmetric matrices in the book

Further Reading · 12.1 The Data Matrix

From geometry toward data science — where eigendecomposition gets applied

Open-source Code Repository

All code and figures in the book: <https://github.com/Visualize-ML>

Exercises Q1 – Q9

Q1a By hand, find the eigendecomposition of $\text{diag}(2,2)$ and $\text{diag}(1/2,2)$ from Table 1; explain the geometric difference.

Q1c Write Python code to compute the eigendecomposition of all 18 matrices in Table 1, and check them against your hand calculations.

Q3 Write the characteristic equation $\det(A-\lambda I)=0$ for each matrix in Table 1, and solve it to verify the eigenvalues.

Q5 For each matrix in Table 1, check whether its eigenvectors are linearly independent; if so, V is invertible and A is diagonalizable.

Q7 Use Python to verify $\det(V^{-1}) \cdot \det(V)=1$, and explain whether this holds for any invertible matrix.

Q1b By hand, find the eigendecomposition of $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$; explain why one is diagonalizable and the other isn't.

Q2 For every matrix in Table 1, determine whether it is invertible (is the determinant zero?), and list the determinant values.

Q4 Analyze the eigenvalues in Table 1: which are zero? which are negative? which are complex?

Q6 Prove that for this section's example $A=\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$, both V and V^{-1} satisfy the decompositions $Vx=SKx$ and $V^{-1}x=K^{-1}S^{-1}x$.

Exercises Q10 – Q18

Q10 Write a Python function `asym_evd(A)` that returns V , Λ , V^{-1} , and verify $A=V\Lambda V^{-1}$.

Q12 Use matplotlib to reproduce Figure 5, plotting the intermediate results of the five steps S^{-1} , K^{-1} , Λ , K , S in order.

Q14 Explore: why do the two eigenvectors of the shear matrix $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ end up collinear, making it non-diagonalizable?

Q16 Explain why this section uses `numpy.linalg.eig` rather than `numpy.linalg.eigh` for a non-symmetric matrix.

Q18 Explore: which matrices in Table 1 are symmetric? Verify that their eigenvectors are naturally orthogonal.

Q11 Use matplotlib to reproduce Figure 1, visualizing this section's matrix A acting on e_1 and e_2 .

Q13 Prove that a shear matrix (like K or K^{-1}) always has determinant 1 — it never changes area.

Q15 Construct a 3×3 non-symmetric matrix, find its eigendecomposition with `numpy.linalg.eig`, and check whether V is invertible.

Q17 Prove that the trace (sum of main-diagonal entries) of a non-symmetric matrix A still equals the sum of its eigenvalues.

Exercises Q19 – Q25

Q19 Construct a 2×2 matrix with a zero eigenvalue, and use Python to verify it is indeed rank-deficient (determinant zero).

Q21 Write the matrix-multiplication expression that extracts the entry in row 2, column 1 of V^{-1} .

Q23 Explain why the unit circle, after a non-symmetric transformation A , is still an ellipse (hint: it is still a linear map).

Q25 Preview: the next section introduces the Gram matrix — guess what special property its eigenvalues might have.

Q20 Use Python to verify $A^2 = V\Lambda^2V^{-1}$, and compare the result to computing $A@A$ directly.

Q22 Explore: for a rotation matrix with $\theta=90^\circ$, verify that its eigenvalues are indeed the complex numbers $\pm i$.

Q24 Explore: why is the off-diagonal entry of this section's Gram matrix VTV equal to $-\sqrt{2}/2$? How does it relate to the angle between v_1, v_2 ?

10 Prompts to Explore with AI

- 1 Explain the eigendecomposition $A=V\Lambda V^{-1}$ of a non-symmetric matrix, and how it differs fundamentally from spectral decomposition.
- 2 Explain geometrically why V is not orthogonal here, and why V^{-1} cannot be written as V^T .
- 3 Explain how to use the Gram matrix $V^T V$ to check whether V 's columns are unit vectors and orthogonal.
- 4 Derive why V can be factored into a shear matrix K and a scale matrix S .
- 5 Explain which matrices correspond to each of the five steps in “scale→shear→scale→shear→scale”.

10 Prompts to Explore with AI

6 Explain why the determinants of a matrix and its inverse always multiply to 1, and verify it with this section's example.

7 Explain what geometric transformations a zero eigenvalue and a negative eigenvalue each correspond to.

8 Explain why shear matrices often have repeated eigenvalues and are not diagonalizable.

9 Use Python to demonstrate the eigendecomposition of any matrix in Table 1, and visualize its geometric transformation.

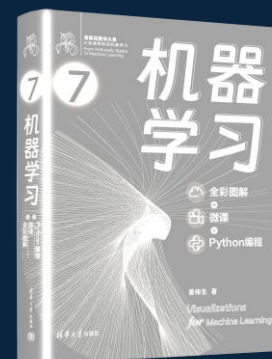
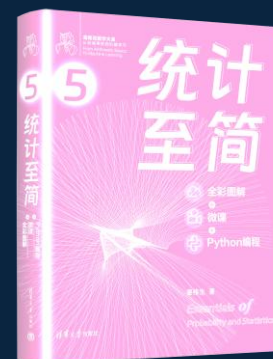
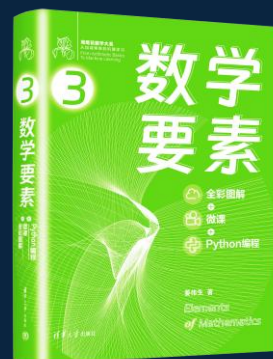
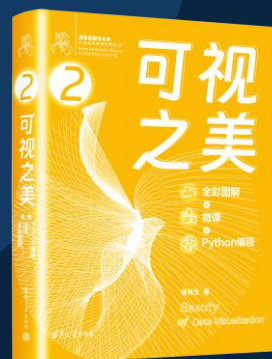
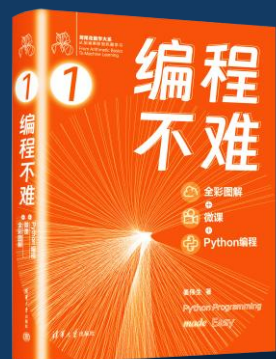
10 Explain why rotation matrices generally cannot be diagonalized over the reals, and when they can.

数学

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